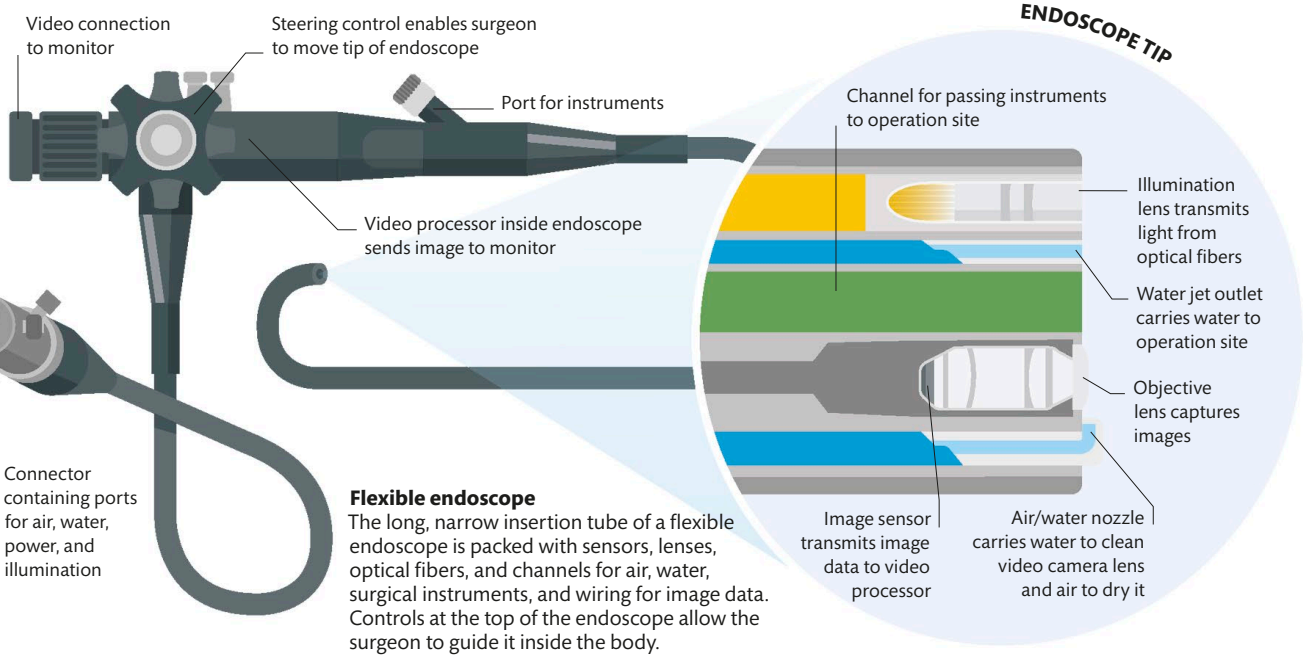
Flexible endoscopy

In this form of surgery, a flexible endoscope is introduced into a body cavity, such as the windpipe or intestine, through the mouth or other natural opening. The endoscope contains optical fibers to transmit light to the operation site and a video camera at the tip to send images from the site back to a monitor. It also has channels to pass air, water, and surgical instruments to the operation site.

10,000
THE NUMBER OF **OPTICAL FIBERS** IN SOME FLEXIBLE **ENDOSCOPES**



Flexible endoscope
The long, narrow insertion tube of a flexible endoscope is packed with sensors, lenses, optical fibers, and channels for air, water, surgical instruments, and wiring for image data. Controls at the top of the endoscope allow the surgeon to guide it inside the body.

Robot-assisted surgery

Some forms of keyhole surgery can now be performed with the help of a robotic system. Robotic arms are mounted on a cart beside the patient. An endoscope on one arm transmits views from inside the body to the surgeon's console and to a video monitor. The other arms hold surgical instruments. The surgeon uses hand controls in the console to move the instruments inside the patient. One of the advantages of robotic surgery is that the robotic system can scale down the surgeon's movements, enabling more precise control of the instruments.

